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GANs beyond Image Synthesis

Project management(GEP)
Final Assignment

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Specialization in Computing

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Context and scope

1.1 Introduction and contextualization

The recent advances in Deep Learning have substantially improved Machine Learning algorithms to the extent that they can defeat humans in some tasks. However, we can not conclude that this algorithms truly understand their input data without checking if they are able to create new one. The most promising approach in this task is the Generative Adversarial Networks, or GANs for short, introduced by Goodfellow et.al in 2014 [3].

Although the most common usage of Generative Adversarial Networks is the generation of new images, this type of generative model can be used in several other areas due to its data modeling property. In this thesis we are going to employ GANs and the adversarial training process to Data Augmentation and Outlier Detection tasks.

The generation of artificial training data can be deeply useful to have a high-quality data set to train algorithms. As it's been studied [13], Data Augmentation can be used to generate a new synthetic training data set, keeping the privacy of the original one. That can be helpful when the original data set is composed by sensitive information that may be mistreated. Furthermore, classification algorithms may have problems identifying the minority classes in imbalanced data sets since they would have a low error even though the minority class is miss classified. Hence, as in real-world we can not expect to work with balanced data sets, Data Augmentation can help us over sample the minority class and GANs may help.

GANs have revealed high qualities in modeling high dimensional distribution of real world data. This capacity of learning the manifold of data

suggests [4] that they can be used for anomaly detection, since they can successfully learn how to map samples from an arbitrary latent distribution to data, understanding the anomalies of the samples. Solving this problem can help to detect network intrusions or bank frauds, among other applications.

1.1.1 Context

This is a Bachelor Thesis of the Computer Engineering Degree, specialization in Computing, done in the Facultat d'Informàtica de Barcelona of the Universitat Politècnica de Catalunya, directed by René Alquezar Mancho and mentored in a external way by the AI department of Basetis, a consulting firm from Barcelona.

1.1.2 Concepts

The reader must be familiarized with the following concepts to understand correctly the work . Moreover, there is a briefly description of the architectures that are going to be used for studying the GANs applications on the mentioned tasks.

Generative Adversarial Networks

The Generative Adversarial Networks [3] are frameworks for the estimation of generative models via an adversarial competition, in which two Artificial Neural Networks (ANN), composed by one discriminator D and one generator G, are trained simultaneously, improving both models at the same time and aiming to reach the Nash equilibrium¹ of costs [15].

The goal of the generator G [13] is to capture the data distribution, mapping a latent² space to the distribution of the data it aims to reproduce. As we will observe in the later section, one of the problems that GANs have to deal with in anomaly detection is learning how to map the data back to the latent space (the inverse mapping) during training time. The discriminator D evaluates the quality of the data created by the generator, receiving as input data samples from either the generator or from the original data set, and trying to predict the source of the sample. As we are training both networks at the same time, they will improve while competing against each

¹Studied in Game Theory, a Nash equilibrium happens when the players can't perform better even if they change their strategy.

²A latent space is a representation of compressed data in which similar data points are closer together in space

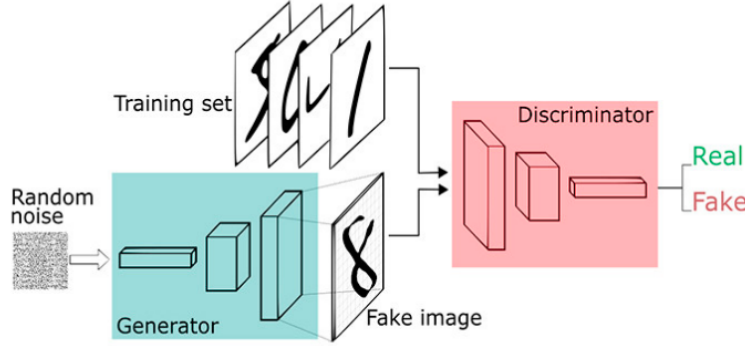


Figure 1.1: Generative Adversarial Network [3].

other. The discriminator model D will try to get better at distinguishing between the real data and the data generated by G , and the generator G will get better at generating data closer to the original.

As Goodfellow stated [3], we can think the adversarial framework as a minimax game that has a unique solution, the one with the generator model G being able to represent correctly the training data distribution and the discriminator D having an accuracy of $1/2$. We train D to maximize the probability to label correctly the training examples and the samples of G , and we train simultaneously G to minimize $\log(1 - D(G(z)))$. One way to prevent the discriminator of being trained much better than the generator is training it to maximize $\log(D(G(z)))$ instead of minimize $\log(1 - D(G(z)))$ [6], since when D rejects with a confidence close to 1, G can not learn from zero gradient. See 1.1.

$$\min_G \max_D V(D, G) = E_{x \sim p_a(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (1.1)$$

State-of-the-art algorithms for anomaly detection based on GANs

Federico Di Mattia, Paolo Galeone, Michele De Simoni and Emanuele Ghelfi have gathered in "A Survey on GANs for Anomaly Detection" [4] the state-of-the-art algorithms for Outlier Detection. AnoGAN, proposed by Thomas Schlegl [15], was the first architecture suggested in this field. It uses a standard GAN, trained on normal samples only, with the purpose of learning how to map the samples to the latent space and vice versa (using the BiGAN

concept), learning the manifold of the normal samples. The underlying idea was to encode an anomalous sample to a normal one on account of spotting the differences (anomalies) between the two. However, the minimization through Γ back propagation steps of the loss function for every new input sample makes unfeasible the test-time performance.

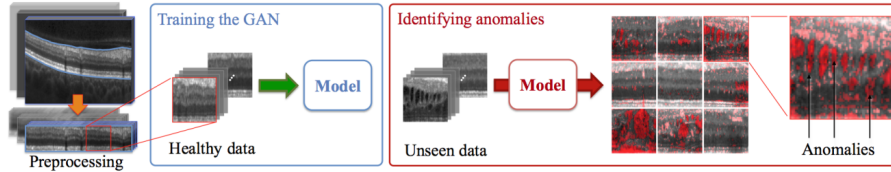


Figure 1.2: AnoGAN [15].

The Efficient GAN-Based Anomaly Detection (EGBAD), proposed by Zenati [2], tries to solve the AnoGAN disadvantages using the BiGAN architecture³. Hence, its main contribution is allowing to compute the anomaly score without the Γ optimization steps during the inference as it happens in the AnoGAN case.

The GANomaly architecture, introduced by Akcay in 2018 [12], trains a generator network on normal samples to learn their manifold⁴. It is intended to improve other approaches to the anomaly detection problem, like AnoGAN [15] (that has huge performance issues) and is inspired by EGBAD [2]. The generator network of GANomaly is composed by three elements: an encoder G_E , a decoder G_D and another encoder E . As the encoder is learned during the training process, there is not need for the iterating process so it makes the learning process faster.

This three GANs architectures are the state-of-the-art in Outlier Detection. Even though we will go in further details in later chapters, they are worth to mention in the introductory concepts.

³The BiGAN architecture includes an encoder E that learns the inverse of the generator $E = G^{-1}$, allowing to also learn a mapping from data to latent space.

⁴A manifold is an abstract mathematical space where every point in it has a neighbour that resembles an Euclidean space. The idea is to map the points in the higher dimensional input to the lower manifold.

State-of-the-art algorithms for data augmentation based on GANs

Despite the fact that we are going to approach data augmentation to standard numerical data sets and that there aren't models explicitly designed to them, it is worth to study the image synthesis models [13]. However, there are a few differences that we will have to take into account, such as the non appliance of convolutional layers, since there are no positional relationships over the sample vectors.

Many approaches, such as Variational Autoencoders (VAEs)⁵ or autoregressive models, intend to build image synthesis models but non of them are able to provide a latent representation of data and they are costly to sample. ACGAN [1], proposed in 2017, modifies the standard GAN formulation to introduce a class label to each sample, in addition to the noise. The discriminator will give both a probability over the class labels and over the sources, and the generator will use both to generate images. The objective function is composed by the log-likelihood of the correct source L_s , and the log-likelihood of the correct class L_c [1]. The discriminator model is trained to maximize $L_s + L_c$ (1.2) and the generator to minimize $L_c - L_s$. One of the main contributions of ACGAN is that it permits the separation of data sets by classes and it trains a generator and a discriminator for each one.

$$L = E[\log P(S = real | X_{real})] + E[\log P(S = fake | X_{fake})] \quad (1.2)$$

BAGAN, proposed in 2018 by Giovanni Mariani et.al [5], use an autoencoder to initialize the GAN towards a good solution, initializing G with the weights of the decoder, and the first layers of D with the weights of the encoder E. Hence, it first trains the autoencoder, and then transmits its knowledge to the GAN for the adversarial training, optimizing the parameters in G to match the labels used to generate the images.

1.1.3 Problem to be solved

Although the large majority of the Generative Adversarial Nets uses are related to image generation, the last few years this framework has been employed in other tasks. In this thesis we want to check if this architectures can contest its competitors in Data Augmentation and Outlier Detection

⁵Autoencoder that regularizes its encoding distribution to ensure good properties of the latent space, being able to generate new data.

problems in real-world situations, studying the state-of-the-art and comparing the GANs architectures with the principal contenders. As stated before, Data Augmentation may have a crucial impact in Machine Learning models performance, and we also want to check if GANs can have a positive footprint in the outlier detection field.

1.1.4 Stakeholders

Many parties involve this project, in a direct or indirect way, depending on the type of benefit or influence drawn from the work. The **tutor**, René Alquezar Mancho, the AI department of Basetis, concretely the **head of the AI department** Martin Goethe, and the **researcher**, are stakeholders that have a direct implication on the thesis. Both the tutor and the head of the AI department will mentor the researcher, that is researching as an intern for the AI team of the consult firm.

Secondly, the stakeholders that are indirectly involved on the thesis are the ones that take the most out of the work. If we can check that the GANs architecture has a solid performance in the mentioned tasks, all the companies may include this framework in their work methodology, and It can encourage other researchers to invest their time in this area.

1.2 Justification

1.2.1 Previous studies

Considering that Generative Adversarial Nets were proposed in 2014 [3], there has been an immense growth of research in this field over the last few years. Although the most studied application is image synthesis [6], in this thesis we will focus to study this framework in the Outlier Detection and Data Augmentation tasks beyond image generation. The first architecture proposed using GANs for outlier detection was AnoGAN [15], introducing an anomaly scoring scheme based on the mapping from image space to latent space. Zenati et.al [2] introduced the BiGAN concept into the EGBAD model, solving the minimax optimization problem during training time, saving the time spent on the Γ optimization steps during inference as in AnoGAN. Employing two encoders and one decoder, GANomaly [12], proposed in 2018 by Samet Akcay et.al, it's been proven to be the superior model in the field.

In data augmentation domain we have few architectures regarding standard numerical data bases. However, as It's stated in "Data augmentation using GANs" [13], the techniques used are similar, so they are worth to explore. Odena et.al [1] propose a variant of the GAN architecture, ACGAN (Auxiliary Classifier - GAN), that permits the classification of data set into subsets by class and trains a generator and discriminator for each one. BAGAN, proposed in 2018 [5], uses the class conditioning concept from ACGAN [1] to bring class knowledge to the model and improves it using an auto-encoder based initialization and modifying the discriminator output topology.

1.2.2 Justification

As seen in the preceding paragraph, GANs are achieving an impressive performance in fields such as Computer Vision and Natural Language Processing and its uses are deriving beyond image generation. Having an efficient and optimal model in Data Augmentation and Outlier Detection problems will help companies to make a huge step forward, allowing them to perform better in their models, keeping the privacy of sensible data sets and building better models in areas such as fraud or illness detection.

1.3 Scope

1.3.1 Objectives and sub-objectives

The generic objective of this thesis is gather together all the promising state-of-the-art GANs architectures and apply them in what we consider two important fields of interest, Data Augmentation and Outlier Detection, explore its potential by making a comparison among its competitors and observe if they can really have an impact in real-world situations. To accomplish the objective, we can subdivide the project in numerous sub-objectives.

Theoretical Part

- Study the GANs framework.
- Explore the described architectures for Anomaly Detection and Data Augmentation.
- Explore the alternatives to GANs for each task and research over the main competitors.

- For each method
 - Compute the spatial and time complexities.
 - Bound the generator loss.
 - Analyse the anomaly score (Outlier Detection).
 - Research on Pros and Cons.

Practical Part

- Program the methods studied under the same framework (Tensorflow) and within the same code-base.
- For Data Augmentation
 - Decide a classifier to test the generated/augmented data set over the original data set.
 - Check the performance of a full synthetic data set.
 - Check the performance of a data augmented data set.
- Compare the results obtained with the ones computed in the theoretical part.
- Draw conclusions concerning the results.

1.3.2 Requirements

Some requirements are needed to ensure the quality of the thesis

- Select the optimal parameter and hyper-parameter configurations of the ANNs.
- Add an evaluation process to make model selection and select the best model for each architecture during training period.
- Data sets used for Data Augmentation must have a notable imbalance.
- Choose a visual classifier to test the Data Augmentation generated data sets.
- Certain need of high computing power to train the Generative Adversarial Networks quickly.
- Define the comparison methods that are going to be used.
- Use good programming practices, with a readable style and least complexity possible.

1.3.3 Potential obstacles and risks

Throughout the development of the thesis there are some potential risks and obstacles that we may have to deal with.

- **Deadline of the project.** There is a delivery deadline that needs to be achieved, hence, It conditions the longevity of the thesis so we must define accurately the distribution of the work to avoid haste.
- **Inexperience in the field.** It's the first time I'm getting involved in a Tensorflow project and also the first I'm researching in Deep Learning area, that may pull myself back.
- **Computational power.** The Generative Adversarial Networks are computing expensive architectures and If we can't guarantee enough computing power that may delay the work.
- **Coronavirus.** Working on a thesis while there is a global pandemic has its consequences, there is the risk of getting confined, making impossible the physical meetings with the tutor or the use of external computing power.

1.4 Methodology and rigor

Defining correctly the working methodology that is going to be followed, the monitoring tools and the validation methods is decisive to the proper development of the thesis.

1.4.1 Methodology

Given that the thesis has to be completed in a short period of time, approximately 4 months, we have opted to inspire ourselves with the *agile* methodology, that subdivides the global work into sub tasks or *sprints* of two weeks. In my case, due to the fact that I'm studying a lot of different architectures over two different problems, Data Augmentation and Outlier Detection, each sprint (longevity to be defined) may be the study and experimentation of one architecture. So I will set aside one *sprint* for each architecture, to then analyze the results and make the comparison among them.

1.4.2 Monitoring tools and validation

We will use a GitHub repository as a tool for version control due to the easy access (cloud storage) and the failure recovery, since it stores previous versions of the code. One *master* branch will include the tested code and one *development* branch will comprise all the code that is in development or testing stage. The code will be publicly available to both the tutor and the Basetis AI department, so they all have the possibility to follow the work and the results at any time.

A weekly meeting will be scheduled with the director of the AI department of the consulting firm, and every two weeks a face-to-face meeting with the tutor will be arranged, both with the intention of discussing the project status and checking the tasks to be accomplished the following weeks.

Temporal planning

2.1 Description of the tasks

Owing to the fact that I will perform 360 hours in Basetis, researching and working for the consulting firm, and that in average I expect 8 hours of personal work per week, the total amount of hours committed to the project will be approximately 500. Accordingly, I have to prepare the number of hours dedicated to each task to ensure the good performance of the thesis. The work-to-begin date is on September 14th and the delivery date on the last week of January, so I have to distribute 500 hours of work among 129 days.

2.1.1 Task definition

This section shows the tasks that will be done throughout the project. Moreover, each task is described taking into account the dependencies with other tasks and the duration of each one. We gathered the tasks in groups that are written in work order. Therefore, there are **time dependencies** between each group of tasks, so to start working on the *Practical implementation* we need to finish the *Study of the state-of-the-art* group of tasks and so on.

Project planning

Given the importance of the planning in the final work, we present all the tasks that will be done over project management, on account of defining and documenting the work to perform.

- **Contextualization and project scope.** Definition of the project scope in the context of its study. Indicating the objective of the thesis, the relevance of the area and how it is going to be developed.

- **Temporal planning.** Planning of the entire execution of the work, providing a description of the phases, resources and requirements needed.
- **Economic management and sustainability.** Analysis of the sustainability and the economic dimension of the project.
- **Meetings.** Meetings with the tutor will be scheduled every two weeks with the purpose of ensuring that the thesis is evolving correctly and fulfilling the time plan.
- **Integration in final document.** Bring together all the tasks, correcting the parts that were mistaken.

Study of the state-of-the-art

Considering that one of the main purposes of the thesis is to research over the state-of-the-art GANs architectures in Data Augmentation and Outlier Detection problems, it is fundamental to have a solid theoretical part to ensure the consistency of the experiments. However, before I start working with the different architectures, one important task is to **familiarize myself with the Generative Adversarial Frameworks**. As it is mandatory to study the architectures extensively, we are going to perform the following tasks for each one.

- Describe the architecture and how it is adapted to the problem (e.g. including an extra encoder or the BiGAN structure).
- Compute the spatial and time complexities
- Bound the generator loss theoretically.

Practical implementation

In order to compare the mentioned architectures in both problems, we have to program each one of them. Even though we may find some libraries that ease our work, since the main purpose of this thesis is doing research over the GANs, it is convenient to program each architecture from scratch to have more freedom to apply modifications.

We divide the practical implementation in the following tasks.

- **Learn Tensorflow.** Using this language to implement the networks makes mandatory familiarizing myself with it before starting to write code.

- Program the GANs applied to Data Augmentation
- Program the GANs applied to Outlier Detection
- Test the correct functioning of the programmed networks to ensure the correct functioning of each one.

Experimentation, analysis and conclusion

Providing a good analysis and a coherent conclusion is one of the most important tasks of a thesis. For instance, it will miss the point if we don't finish this last part correctly, so the following tasks are crucial for the proper functioning of the project.

- **Select the data sets.** Given that we have to choose data sets for Data Augmentation and Outlier Detection problems, we have to find imbalanced data sets. In the case of Outlier Detection we can consider the imbalanced class as the anomaly class.
- **Select the benchmarks.** It is important to have a standard point of reference to compare the architectures in order to avoid bias.
- **Experimentation.** After finding the optimal parameters for the architectures using a model validation technique (e.g. cross-validation) we have to experiment with each architecture over the data sets, looking to the further analysis.
- **Conclusion.** Analyze the results found in the experimentation part and make the conclusion.

One implicit task that needs to be done throughout the thesis is the **documentation**. It is important to document each task to avoid the haste of the final weeks. The last task of the work is the **preparation of the oral defense**, considering the important parts of the thesis that need an explanation.

2.1.2 Summary of the tasks

In the following table I summarized the previously explained tasks, showing the dependencies among them and including also the amount of hours dedicated to each one.

Id.	Task	Time(h)	Dependencies
T1	Project planning	35	-
T2	- Contextualization and project scope	10	-
T3	- Temporal planning	7.5	-
T4	- Economic management and sustainability	7.5	-
T5	- Integration in final document	10	-
T6	Meetings	5	-
T7	Study of the state-of-the-art	200	-
T8	- Familiarization with GANs	50	-
T9	- Description of the architecture	50	T8
T10	- Computation of the spatial and time complexities	50	T8,T9
T11	- Bound the generator loss	50	T8,T9
T12	Practical implementation	160	T7
T13	- Learn Tensorflow	30	-
T14	- Program the GANs applied to Data Augmentation	60	-
T15	- Program the GANs applied to Outlier Detection	60	-
T16	- Test the programmed networks	10	-
T17	Experimentation, analysis and conclusion	100	T12
T18	- Select the data sets	20	-
T19	- Select the benchmarks	10	-
T20	- Experimentation	40	-
T21	- Conclusion	10	-
T22	Documentation	30	-
T23	Oral exposition	10	T17

The amount of hours dedicated to each task is approximated, I expect to remake the project planning after a few weeks of work to make it more consistent.

Table 2.1: Summary of the tasks.

2.1.3 Resources

To perform well on the thesis there are some resources, human and material, that I need to count with.

Human resources

The main human resource of the thesis is the researcher, since is the one working on it. Both the university tutor René and the Basetis tutor Martin will mentor the researcher on its task, so they are also fundamental human resources. Finally, the GEP tutor is in charge of correcting the project man-

agement part, so he will have a big influence in the scope and organization of the work.

Material resources

Considering that research projects are grounded on previous works, there is a certain need for material resources such as books or papers. Furthermore, I will need some software and hardware resources to make the practical part and the experiments.

- **Overleaf.** We will use *Latex* given the easy text formatting and organization, and Overleaf is the best online tool for it.
- **Atenea.** Used to communicate with the professor in charge of GEP and deliver the work.
- **IDE.** A solid IDE (to be defined) is needed to ease the practical work.
- **Github.** Github will be used as a tool for Version Control due to its accessibility and version recovery.
- **Computers.** The experiments will be done with a MSI laptop computer, with 16GB of RAM and Intel(R) Core(TM) i7-6700HQ CPU.

2.2 Gantt

Below (2.1) I attach the Gantt chart where I show how the tasks are going to be distributed among the semester, considering the number of hours of each task (see 2.1.2) and the dependencies described in the previous section. The start day of the project is the first day of the semester, and since I will defense the thesis in the first call, the end date will be on the last week of January.

The meetings with tutors will be scheduled in function of my necessities, so even though there will be one control meeting every two weeks, some weeks I will need more. Hence, I considered to include the meetings to the chart after a month or two of work.

2.3 Risk management

Some risks may appear through the course of the project and influence negatively its progress (see 1.3.3). In this section, I will present the potential

risks, how they would affect the project and how can I solve them via introducing alternative tasks.

2.3.1 Deadline of the project

One of the problems of planning before starting the project is that I may underestimate the number of hours dedicated to each task. Hence, that may end up distorting all the project plan, so I must know how to react to that situation.

- **Impact:** Medium
- **Proposed solution:** Adapt the project plan to the new situation and recalculate the hours needed for each task. In case I have to redo the initial time plan and due to the fact I dedicated one week to it (approximately 20 hours), I shall create another task with 20 hours of estimated dedication to solve the problem. What seems to be a waste of time, it is critically important to ensure the finalization of all the parts of the project.

2.3.2 Inexperience in the field

Given that Generative Adversarial Networks is a new domain for me, reaching the sufficient level of being capable of implement state-of-the-art architectures may be more time demanding than is expected.

- **Impact:** Low
- **Proposed solution:** Dedicate more hours to study GANs. Although I have reserved the first three weeks not only to study GANs theoretically, but also to get in touch with Keras and Tensorflow, there is the potential risk of under reaching the required knowledge. If I see myself at the limit, I can reduce the number of hours per week dedicated to *Description of the architecture* task (approximately all November, see 2.1) and include some hours to study more deeply GANs and the mentioned libraries.

2.3.3 Computational power

Generative Adversarial Networks are a computing expensive framework. Therefore, It may happen that in architectures that are even more computing demanding like AnoGAN [15], the training cost will be unfeasible for

a normal computer. In my case, one of the two laptops I'm going to work with hasn't got GPU, so that makes impossible the training over large data sets, so in case I decide to train the models in local, I have to rely on my MSI laptop.

- **Impact:** High
- **Proposed solution:** Explore online alternatives like Google Collaborative or Kaggle Kernel. Even though it has a high impact on the thesis, the solution is quite easy to couple, since the mentioned solutions are free to use and have enough GPUs to train big models.

2.3.4 Coronavirus

In case of another confinement all the work would be done online. It would be the first time I will have to work from home so that may decrease my productivity.

- **Impact:** Low
- **Proposed solution:** Adapt myself and prepare a good place to work where I feel comfortable. If it is necessary, use part of the contingency budget (see section 3) to buy the required equipment for the proper working environment.

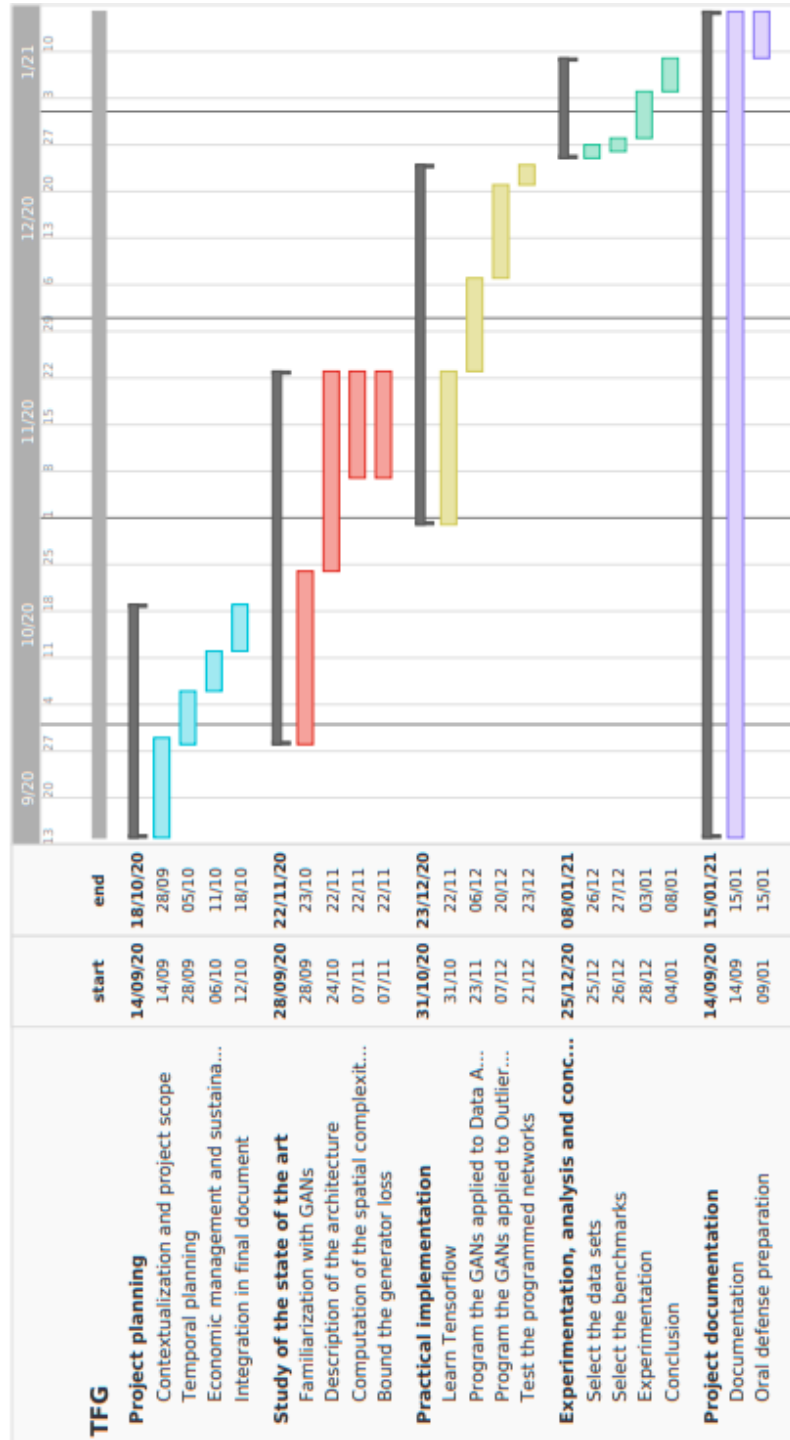


Figure 2.1: Gantt Diagram.

Budget

In this section I will discuss the economic cost of the project. To do so, I will describe the personnel cost, specifying the roles needed, and the generic and indirect costs. Furthermore, I will explain the mechanism for controlling potential budget deviations, including the numerical indicators.

3.1 Staff costs

To make a proper estimation of the personnel costs, I must define which roles are needed, its cost per hour and the distribution of the tasks (described in 2.1.2) among them. The workers salary is calculated by multiplying its hourly wage by the number of expected hours. In addition, the cost of a task is computed by summing the cost of the workers involved.

Although the roles are going to be performed by myself and the tutors, I shall break down the work into them for the best understanding of the global cost. Hence, there are 5 roles that need to be played, each with a specific cost per hour (see 3.1). Firstly, the **Project Manager** is the responsible of the planning and execution of the project. Secondly, the **Junior Researcher** will study the GANs beyond Image Synthesis, describing and experimenting with them, to finally let the **Analyst** draw the conclusions. The **Junior Developer** and the **Tester** are the ones in charge of implementing the GANs and verifying its correct functioning. The Project Management role involves the tutors, but the rest are going to be played by myself.

Given the definition of the roles needed and its cost, the next step is to define the relation between the Gantt diagram (2.1) and the different roles (see 3.1).

Role	Cost(€)/h
Project Manager	23 [7]
Junior Researcher	15 [8]
Junior Developer	22 [9]
Tester	8 [10]
Analyst	13 [11]

Table 3.1: Cost per hour of the different roles.

Task	Hours	Project Manager	Junior Researcher	Junior Developer	Tester	Analyst
Project planning	35	35	-	0	0	0
Meetings	5	5	0	0	0	0
Study of the state-of-the-art	200	0	150	50	0	0
Practical implementation	160	0	0	150	10	0
- Learn Tensorflow	30	0	0	30	0	0
- Program the GANs applied to Data Augmentation	60	0	0	60	0	0
- Program the GANs applied to Outlier Detection	60	0	0	60	0	0
- Test the programmed networks	10	0	0	0	10	0
Experimentation, analysis and conclusion	200	0	0	50	50	100
Documentation	30	0	15	0	0	15

Table 3.2: Estimated time per task.

Task	Cost(€)
Project planning	805
Meetings	115
Study of the state-of-the-art	3.350
Practical implementation	3.380
- Learn Tensorflow	660
- Program the GANs applied to Data Augmentation	1.320
- Program the GANs applied to Outlier Detection	1.320
- Test the programmed networks	80
Experimentation, analysis and conclusion	2.800
Documentation	225

Table 3.3: Estimated cost per task.

Lastly, in order to define the global cost of the staff, we have to multiply the cost per hour by the number of hours of each employee. In the table below (3.1) I show what would be the cost of recruitment, known as CPA.

Role	Hours	Cost(€)
Project Manager	40	920
Junior Researcher	165	2.475
Junior Developer	250	5.500
Tester	70	560
Analyst	115	1.495
Total	640	10.950

Table 3.4: Total cost of the staff.

3.2 Generic costs

Amortization of the resources

To describe completely the economic cost of the project, I shall take into account the amortization of the resources used. However, owing to the fact that all the software used for the thesis are open source, I am going to focus on hardware amortization (see ??).

I will work approximately 3.6 hours per day during 129 days, 50% in a Toshiba laptop and 50% in a MSI laptop. The formula to compute the amortization is the following one (3.1).

$$Amortization = Resourceprice \times \frac{1}{Yearsofuse} \times \frac{1}{Daysofwork} \times \frac{1}{Hoursperday} \times hoursused(3.1)$$

Hardware	Cost(€)	Life expectancy (years)	Amortization (€)
TOSHIBA laptop	1.500	5	174,41
MSI laptop	650	3	125,96
Total	2.150	-	300,37

Table 3.5: Amortization of the hardware used.

Indirect costs

To be realistic about the budget, I shall identify all the indirect costs since they have a certain influence on the global performance of the work. Even though I can use the Basetis office at La Pedrera, we are in the middle of a global pandemic so half of the week I will be doing tele-working. Thus, I will describe the cost regarding to the expenses of working from home and the ones of travelling to the office.

- **Internet cost:** The internet cost is around 90€ per month. Hence, the total cost would be $(90\text{€}/\text{month}) * (5 \text{ months}) * (11 \text{ days}/30) * (3.6 \text{ hours}/24)$.
- **Travel cost:** I'm using the T-Jove card for the public transport and it costs 80€ per three months. However, It is unavoidable the need of a second card due to the longevity of the project.
- **Electricity cost:** The kWh price[14] is 0.1199 €/kWh. Given that the power of an average desktop is 400kwh, the total cost would be $(0.1199 \text{ €/kWh}) * 400\text{kWh} * 129 \text{ days} * 3.6\text{h} * 0.5$.

Generic cost of the project

Below I summarize the generic cost (CG) of the project, including the amortization of the hardware resources and the indirect costs.

Description	Cost(€)
Amortization	300,37
Internet	24,75
Travel	160
Electricity	86,32
Total	571,44

3.3 Deviations of the budget

Contingency

As in every project, unexpected events may appear during the course of the work so I shall stay one step ahead. Therefore, a contingency fund must be prepared in order to avoid delays in the plan. For instance, there will be an inevitably increment of the PCA due to the blindly distribution of the work among the roles and the contingency fund will help to soften the blow.

Some of the cost are more prone to be incremented than others. Namely, the staff cost is susceptible to changes due to the fact that delays on the planned hours suppose an augment of the workers salary. For this reason, I include a 10% contingency margin to the PCA cost. On the other hand,

due to the time limit of the project, the generic cost is less susceptible to variations, so I define a 5% contingency margin to the GC.

Incidental costs

Section 1.3.3 defines all the possible risks that may appear. In the following table (3.3) I'm going to quantify them in order to include the cost of applying alternative plans into the final budget.

Incident	Estimated cost(€)	Risk(%)	Cost(€)
Deadline of the project (30h)	450	25	112,5
Inexperience in the field (20h)	300	20	60
Computational power	25	50	12,5
Coronavirus	0	70	0
Total	775	-	185

Table 3.6: Incidental costs.

Final budget

The last thing to do is estimate the final budget of our project by putting together all the previous sections. Therefore, in the following table (3.3) I show the total cost of the work.

Activity	Cost(€)
PCA	10.950
CG	571,44
Contingency margin	1.095
Incidental cost	185
Total	12.801.44

Table 3.7: Total cost of the project.

3.4 Management control

Even though I have defined the part of the budget reserved to incidences in the previous section, I haven't described yet how I'm going to model the

potential budget deviations. Below I show the procedure to detect any alteration in the project cost.

While doing each task described in the planning section, I will calculate its individual deviation from the estimated cost. To do so, I will use the following formulas.

- **Estimated cost:** Estimated cost of the task (see 3.1).
- **Real cost:** Real cost of the task. Hence, I shall recalculate the *PCA*, *CG*, *contingency* and *incidents* for each one to check any incidences that may increment its cost. Doing so helps to identify which part of the task has been miss-estimated in order to re-plan it.
- **Deviation:** Difference between the estimated cost and the real cost of the task.

Therefore, the deviation factor indicates how much the real cost varies from its estimation. Note that if the difference between the estimation and the real cost is positive is due to an overestimation of the task cost. I may reallocate this extra money for other incidences. On the other hand, if the difference is negative, I must assign part of the contingency fund to the task in order to cover the deviation.

Sustainability

4.1 Self assessment

The modern world, harmed by the human selfishness, needs urgently a change. The increment of pollution is aggravating global warming, risking the life on Earth in a short period of time. Hence, it is a must to check the footprint of a project. Nevertheless, the new technologies, among other factors, are helping social change and the empowerment of the minorities so it is also important to see the social impact of the work. This two elements, together with the economical assessment, compose the sustainability of the project.

Assessing the sustainability in the Bachelor Final Project not only ensures that the students consider the footprint that their work may have, but also its social and economic impact. Although most students may run ahead of this section, the simple fact of answering the poll and the questions below, makes them take it into consideration when working on their project, evaluating if its implementation is worth for the environment, the society and the economy.

To be honest, I didn't know I was supposed to take into account that amount of indicators when trying to perform a sustainable work. There were concepts like amortization or costs that I had to refresh, important to understand the impact of the work in all the different areas. When thinking about sustainability, the first thing that comes to my mind is the social impact, like equity, and the environmental impact it has. However, I used to omit the economical area even though it is critical to check if the project is economically viable.

4.2 Environmental impact

Regarding the PPP, have you thought about the environmental impact of your project? Have you considered minimizing this impact, by for example reusing resources?

Owing to the researching nature of the thesis, it is hard to estimate its environmental impact. As far as I'm concerned, the only point that would have some impact in the environment is the energy consumption due to the training of the different models. However, I have planned with my tutors to train the different architectures in relative small data sets to lower the time needed, thus, the environmental footprint.

Regarding the live expectancy, how is it solved the problem you are trying to solve? Does your solution provide any improvement in the environmental impact?

The Generative Adversarial Networks are computing demanding frameworks, and the different architectures that I'm going to study are even more expensive to work with. Hence, reducing the size of the training sets would make possible its training and will reduce the amount of energy needed. In the [4] paper, the authors didn't experiment with AnoGAN [15] since it needs so computing power that makes the training impossible. They describe it theoretically in order to do the comparison with its competitors. Therefore, I may omit this architecture in order to lower the amount of energy needed.

4.3 Economy

Regarding the PPP, have you estimated the impact that your project will have, including both human and material costs?

In the section 3, the reader will found described the impact of this project, considering both human and material costs, and including the potential deviations that may appear during the course of the work.

Regarding the life expectancy, how is it solved the problem you are trying to solve? Does your solution provide any improvement economically?

Generative Adversarial Networks is the most promising field of Machine Learning and its applications are not yet fully discovered. In this thesis I'm

going to research over applications beyond Image Synthesis and, if I prove that GANs have a better performance than its competitors, that will have a certain impact not only in the quality of the products that my company will make, but also in the computing cost of them.

4.4 Social

Regarding the PPP, how do you think this project will enrich you personally?

To be honest, this is the first time I immerse myself in the research world. The freedom of being able to perform your own work frightens the bravest, but it is also a big opportunity to put into practice all the knowledge collected through this years and. This is also the first time I work for a company, thus, my associates will enrich myself with other points of view.

Regarding the life expectancy, how is it solved the problem you are trying to solve? How do you think your solution will improve people's quality of life? Is there a real need of developing your solution?

As a matter of fact, there is a big expectation in Generative Adversarial Networks. The problems that I'm going to tackle with this framework, and the ones that are not studied yet, may be solved more efficiently and effectively. Hence, it is hard to imagine what would be the real world impact if GANs models solve this problems better than the competitors. What is a small reduction in the computing cost in small data sets is a immense reduction in big applications. Therefore, that would lead to a huge decrease in the environmental footprint.

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